

MICROELECTRONICS CIRCUITS AND APPLICATIONS EAEPC304

PRACTICAL FILE

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Experiment-1

Objective: To design and simulate a Wilson current mirror circuit.

Theory: A current mirror is an electronic configuration that replicates a reference current through one or more active components, allowing it to act as a reliable current source or load. It maintains a consistent output current despite variations in load conditions.

The Wilson current mirror, an improved version of the basic two-transistor mirror, employs three transistors to enhance current accuracy, stability, and output resistance, making it suitable for precision applications.

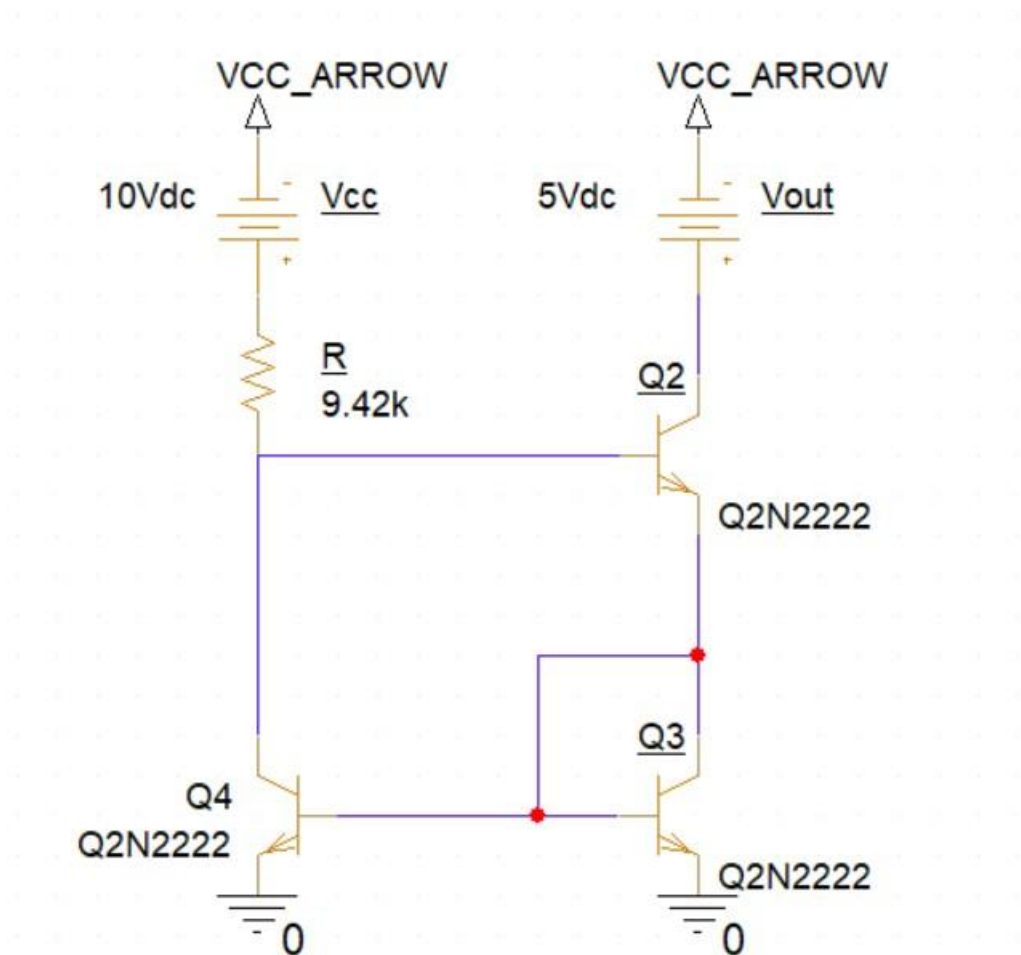
Tools required:

1. OrCAD EDA Suite
2. PSpice Simulator

Procedure:

1. Launch PSpice AD and create the circuit using the necessary components or code.
2. Save the circuit file with a “.cir” extension.
3. Navigate to **File** → **Open Simulation** and select the saved .cir file.
4. Click on the **Run Simulation** icon in the toolbar.
5. Add the relevant traces to the output graph to observe and analyze the resulting waveform.

Circuit diagram:



PSpice Code:

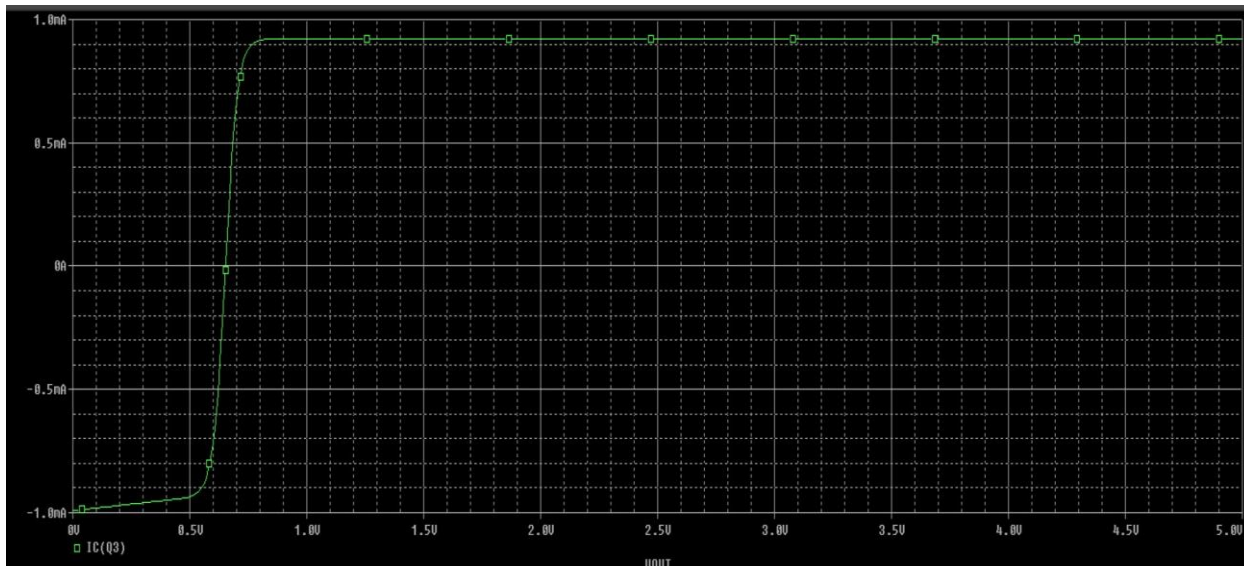
```

1.  ** Cascode NMOS Current Mirror
2.  VDD 1 0 DC 20V
3.  IREF 1 2 DC 3mA
4.
5.  M3 2 2 3 0 MbreakN
6.  M1 3 3 0 0 MbreakN
7.  M4 5 2 4 0 MbreakN
8.  M2 4 3 0 0 MbreakN
9.
10. VOUT 5 0 DC 0
11.
12. .MODEL MbreakN NMOS (LEVEL=1 VTO=1
    KP=5m LAMBDA=0.02)
13. .DC VOUT 0 10 10m
14. .PLOT DC ID(M4)
15. .PROBE
16. .END

```

Result: The Wilson current mirror implemented using Q2N2222 BJTs shows a steady and almost constant output current when the output transistor (Q2) operates in its active region. The collector current of Q2 (shown in green) closely tracks the reference current of Q1 (shown in red), confirming accurate current mirroring once Q2 becomes active.

Output Waveform:



Experiment-2

Objective: To design and analyze a MOS cascode current mirror circuit.

Theory: A current mirror is an electronic circuit used to duplicate a reference current from one active device to another, serving as a constant current source or active load in various applications. It ensures that the output current remains stable despite load or voltage variations.

The MOS cascode current mirror enhances performance by incorporating additional transistors in a cascode configuration. This arrangement keeps the mirror transistors in the saturation region, significantly increasing output resistance and minimizing the impact of voltage changes at the output node, thus providing better current replication accuracy.

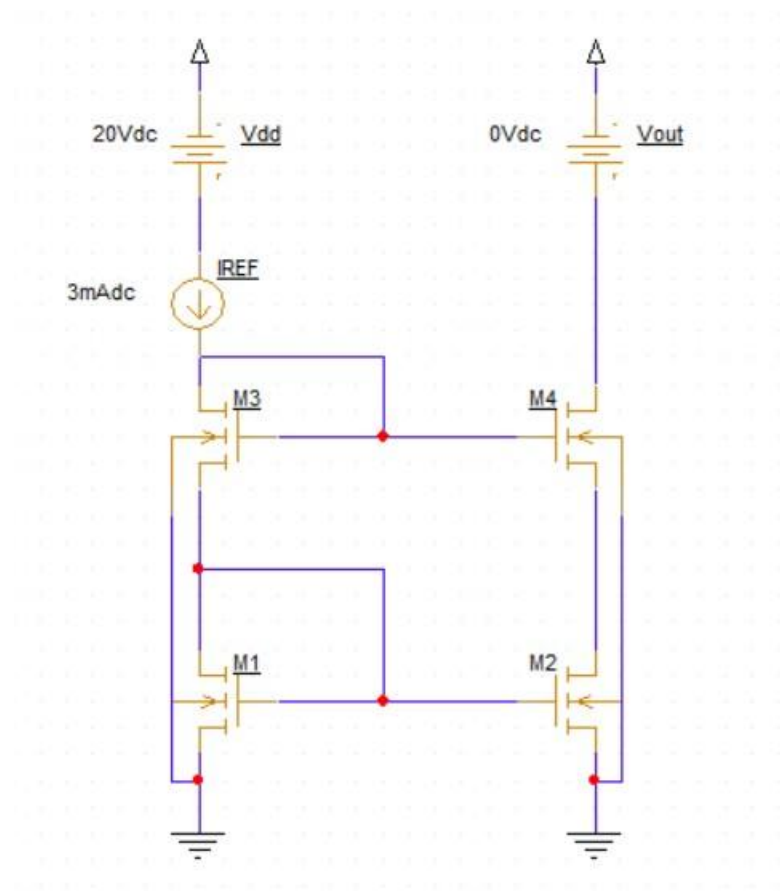
Tools required:

1. OrCAD EDA Suite

2. PSpice Simulator **Procedure:**

1. Open PSpice AD and enter the required circuit code.
2. Save the file in “.cir” format.
3. Go to File Open Simulation and open the .cir file.
4. Click the run simulation icon at the top.
5. Add the desired trace in the output graph to view the output waveform.

Circuit Diagram:

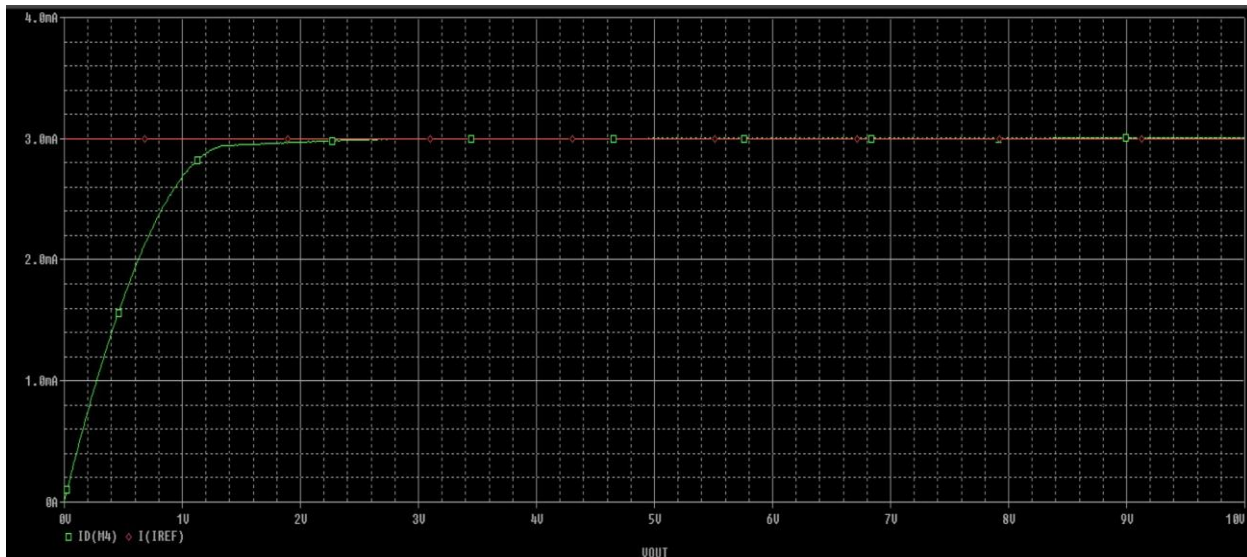


PSpice Code:

1. * Cascode NMOS Current Mirror
2. VDD 1 0 DC 20V
3. IREF 1 2 DC 3mA
- 4.
5. M3 2 2 3 0 MbreakN
6. M1 3 3 0 0 MbreakN
7. M4 5 2 4 0 MbreakN
8. M2 4 3 0 0 MbreakN
- 9.
10. VOUT 5 0 DC 0
- 11.
12. .MODEL MbreakN NMOS (LEVEL=1 VTO=1
KP=5m LAMBDA=0.02)
13. .DC VOUT 0 10 10m
14. .PLOT DC ID(M4)
15. .PROBE
16. .END

Result: The MOS cascode current mirror provides a consistent and precise output current once the transistors operate in the saturation region. Simulation results verify that the output current (through M4) accurately follows the reference current, confirming successful current mirroring behavior.

Output Waveform:



Experiment-3

Objective: To design and simulate a cascode amplifier circuit.

Theory: A cascode amplifier combines a common-emitter stage with a common-base stage to achieve higher gain and improved frequency response. This configuration minimizes the Miller effect, which greatly reduces high-frequency feedback from the output to the input. As a result, the cascode amplifier can handle a wider bandwidth and deliver better stability compared to a single common-emitter amplifier.

Tools required:

3. OrCAD EDA Suite

4. PSpice Simulator **Procedure:**

6. Open PSpice AD and enter the required circuit code.

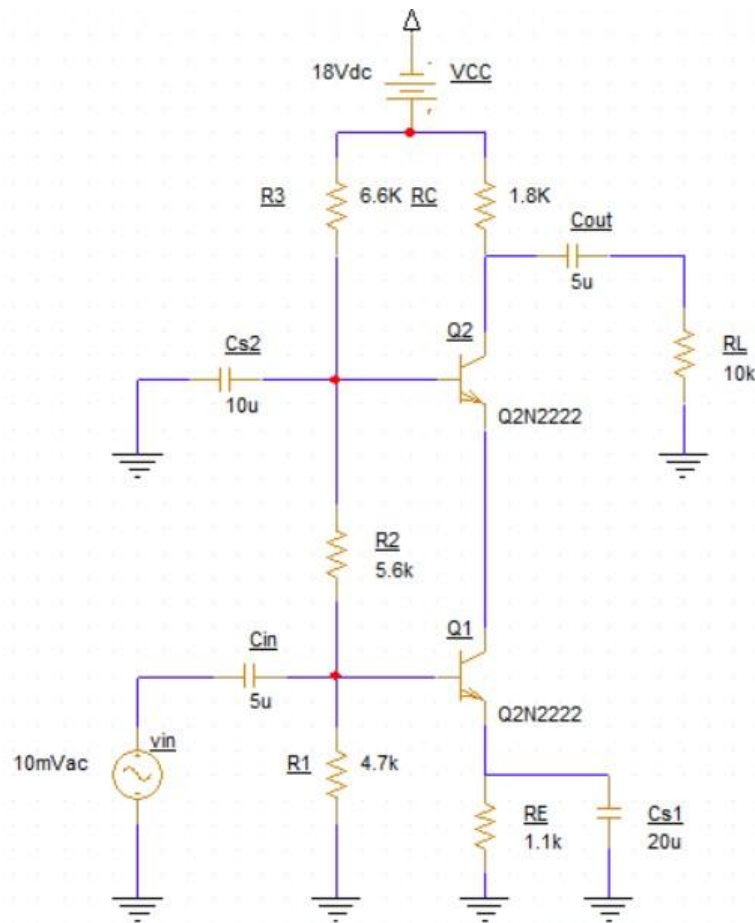
7. Save the file in “.cir” format.

8. Go to File Open Simulation and open the .cir file.

9. Click the run simulation icon at the top.

10. Add the desired trace in the output graph to view the output waveform.

Circuit Diagram:

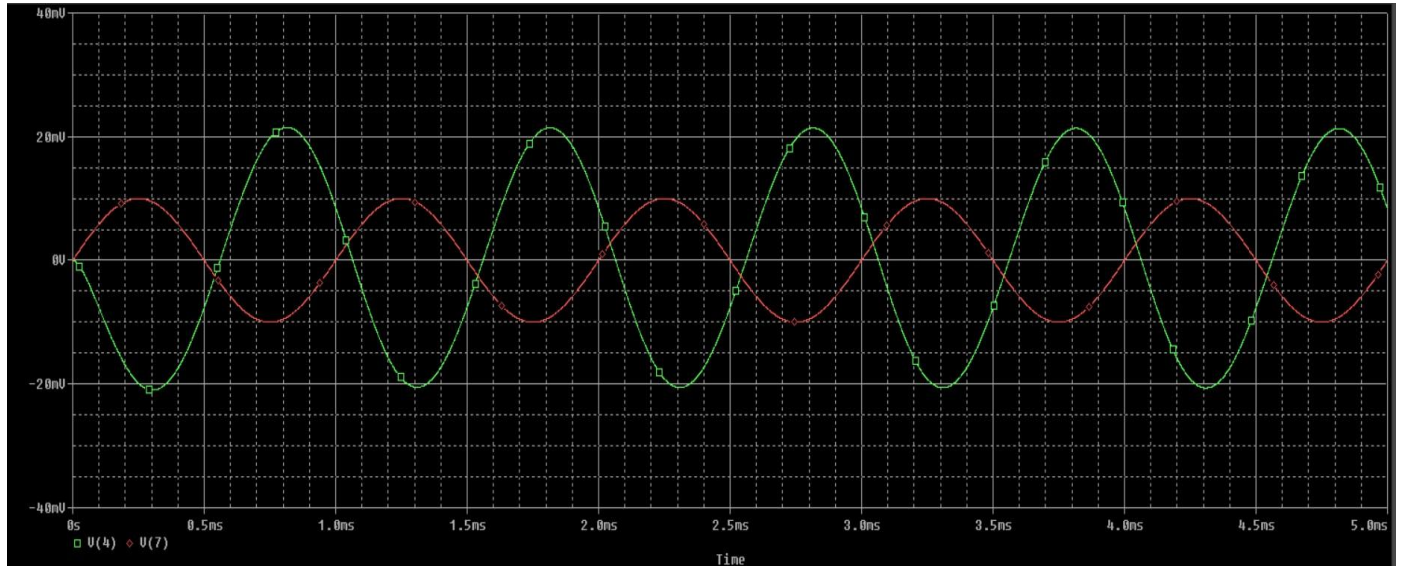


PSpice Code:

1. * BJT Cascode Amplifier	17. Q1 5 6 8 Q2N2222
2. VCC 1 0 DC 18V	18. Q2 3 2 5 Q2N2222
3. VIN 7 0 AC 1 SIN(0 10mV 1k) ;	19.
4.	20. .model Q2N2222 NPN(IS=1E-14 BF=100)
5. R3 1 2 6.8k	21. .tran 0.5u 5m 0 0.5u
6. RC 1 3 1.8k	22. .ac dec 100 10 10Meg
7. Cout 3 4 5u	23. .probe
8. Cs2 2 0 10u	24. .end
9. R2 2 6 5.6K	
10. R1 6 0 4.7k	
11. Cin 6 7 5u	
12. RE 8 0 1.1k	
13. Cs1 8 0 20u	
14. RL 4 0 10k	
15. C1 2 3 10u	
16. C2 4 5 10u	

Result: The BJT cascode amplifier was successfully simulated. The output waveform confirms proper amplification with an approximate 180° phase shift between the input and output signals, validating the expected behavior of the cascode configuration.

Output Waveform:



Experiment-4

Objective: To design, simulate, and evaluate the performance of a CMOS amplifier circuit.

Theory: A CMOS amplifier utilizes complementary MOSFETs — NMOS and PMOS — to achieve high voltage gain with minimal power consumption. In this setup, the NMOS transistor operates in the common-source configuration, providing the primary voltage amplification. The input signal is applied to the NMOS gate, and the amplified output is taken from its drain terminal.

Instead of a resistive load, a PMOS current mirror is used as an active load. This current mirror ensures high output resistance, which enhances voltage gain and maintains a constant bias current through the NMOS transistor. As a result, the circuit achieves stable operation, improved linearity, and efficient amplification.

Overall, the NMOS transistor provides the amplification, while the PMOS current mirror maintains a steady current, resulting in a high-gain, low-power CMOS amplifier with strong performance characteristics.

Tools required:

5. OrCAD EDA Suite

6. PSpice Simulator **Procedure:**

11. Open PSpice AD and enter the required circuit code.

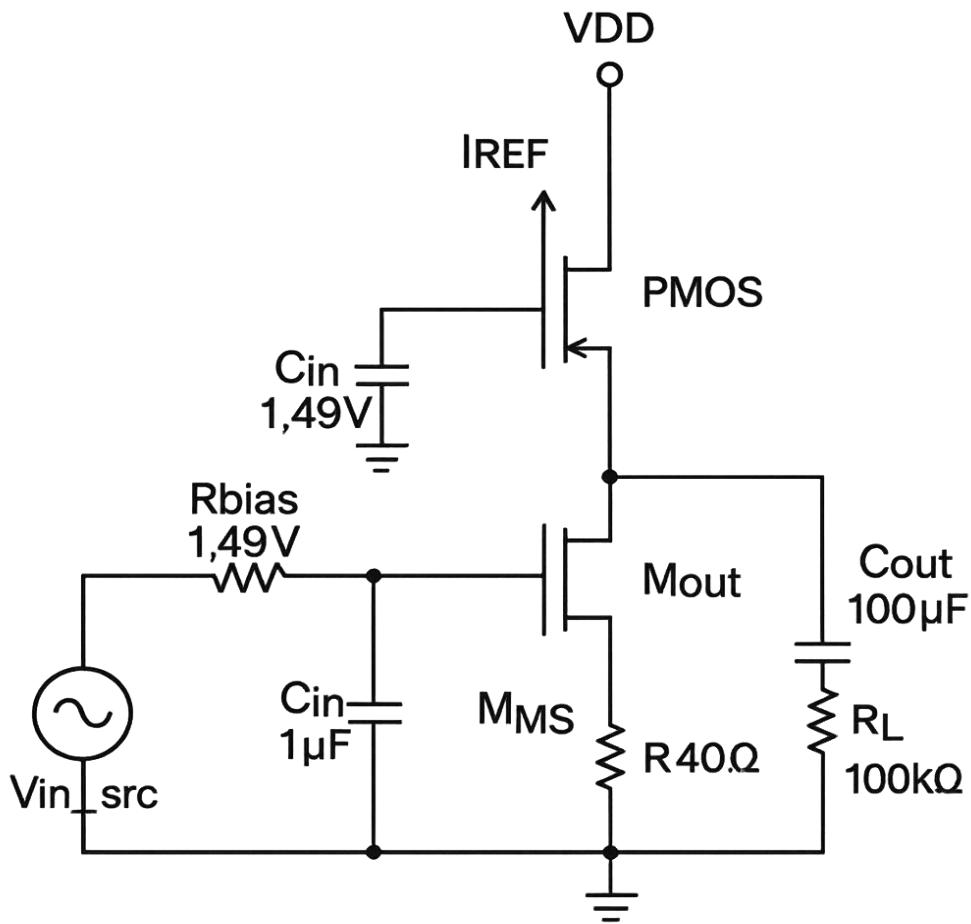
12. Save the file in “.cir” format.

13. Go to File Open Simulation and open the .cir file.

14. Click the run simulation icon at the top.

15. Add the desired trace in the output graph to view the output waveform.

Circuit Diagram:



PSpice Code:

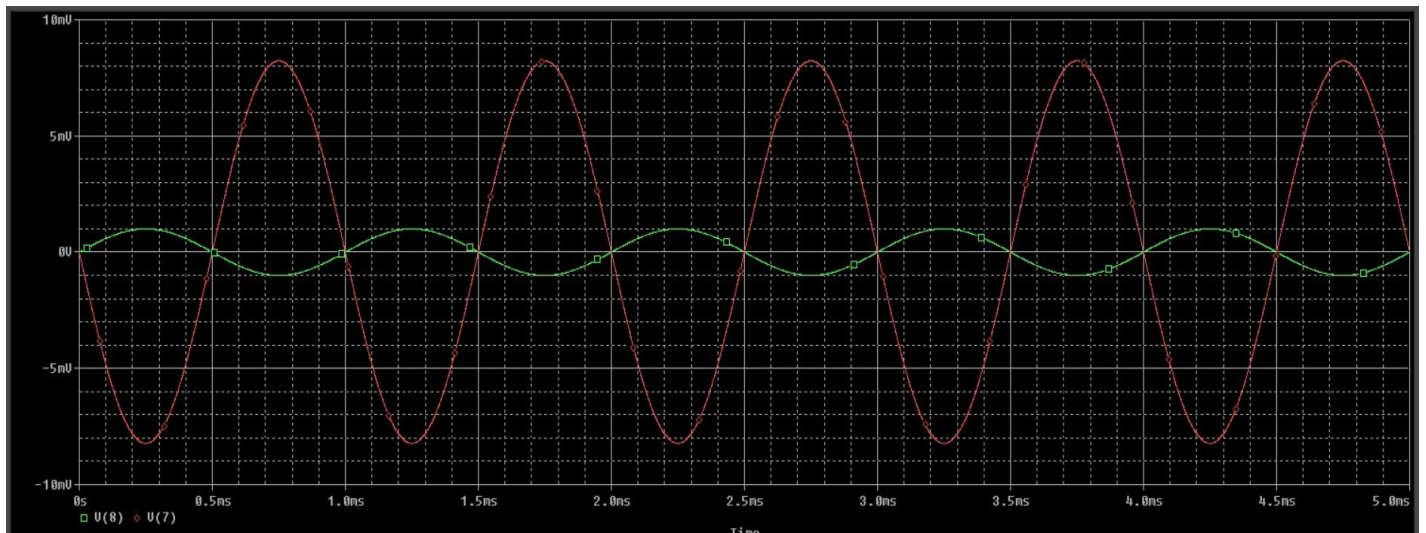
```

1. VDD 1 0 DC 3V
2. IREF 2 0 DC 1m
3.
4. Vin_src 8 0 SIN(0 1m 1k)
5.
6. Vbias 5 0 DC 1.49
7.
8. Cin 4 8 1u
9. Rbias 4 5 1Meg
10.
11. Cout 3 7 100u
12. RL 7 0 100k
13.
14. M1 2 2 1 1 PMOD W=1185.3u L=10u
15. M2 3 2 1 1 PMOD W=1185.3u L=10u
16. M3 3 4 6 0 NMOD W=2390u L=10u
17.
18. Rs 6 0 540
19.
20. .model PMOD PMOS(LEVEL=1 VT0=-0.8
    KP=38u LAMBDA=0.2)
21. .model NMOD NMOS(LEVEL=1 VT0=0.7
    KP=134u LAMBDA=0.1)
22.
23. .op
24. .probe
25. .tran 0.5u 5m 0 0.5u
26. .end

```

Result: The CMOS amplifier successfully demonstrates voltage amplification, with the output waveform showing a clear 180° phase inversion relative to the input signal, confirming proper operation of the amplifier.

Output Waveform:



Experiment-5

Objective: To design and simulate a stable multivibrator using the 555 timer IC.

Theory: An astable multivibrator built with a 555 timer IC generates a continuous square wave output without the need for any external triggering signal. It operates in a self-sustaining or “free-running” mode, where the output alternates between high and low states, producing a steady sequence of pulses.

The circuit's frequency and duty cycle are determined by two external resistors and a capacitor. The capacitor charges through both resistors and discharges through one, causing different charge and discharge durations that define the waveform's timing. Inside the IC, comparators and a flip-flop monitor the capacitor voltage as it varies between one-third and two-thirds of the supply voltage (V_{cc}), switching the output accordingly. Because the operation is entirely automatic and continuous, the configuration is termed *astable*. This circuit serves as a dependable pulse generator, widely used in timing circuits, LED blinkers, waveform generators, and digital clock applications.

Tools required:

7. OrCAD EDA Suite

8. PSpice Simulator **Procedure:**

16. Open PSpice AD and enter the required circuit code.

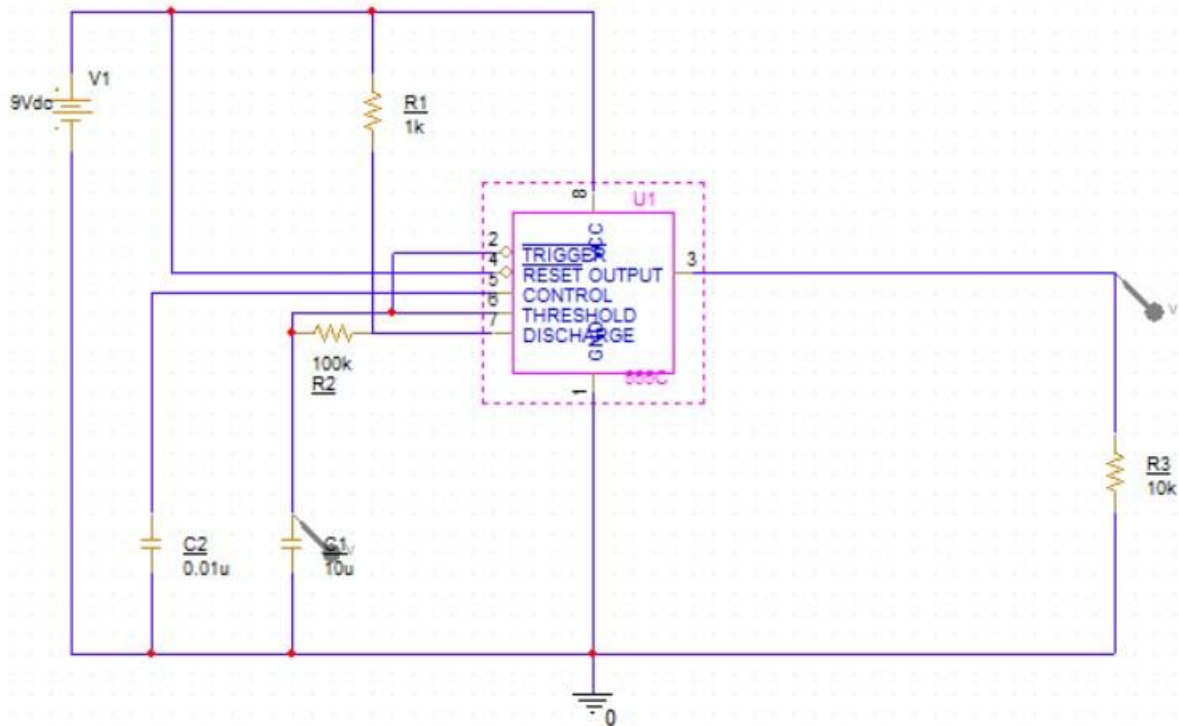
17. Save the file in “.cir” format.

18. Go to File Open Simulation and open the .cir file.

19. Click the run simulation icon at the top.

20. Add the desired trace in the output graph to view the output waveform.

Circuit Diagram:

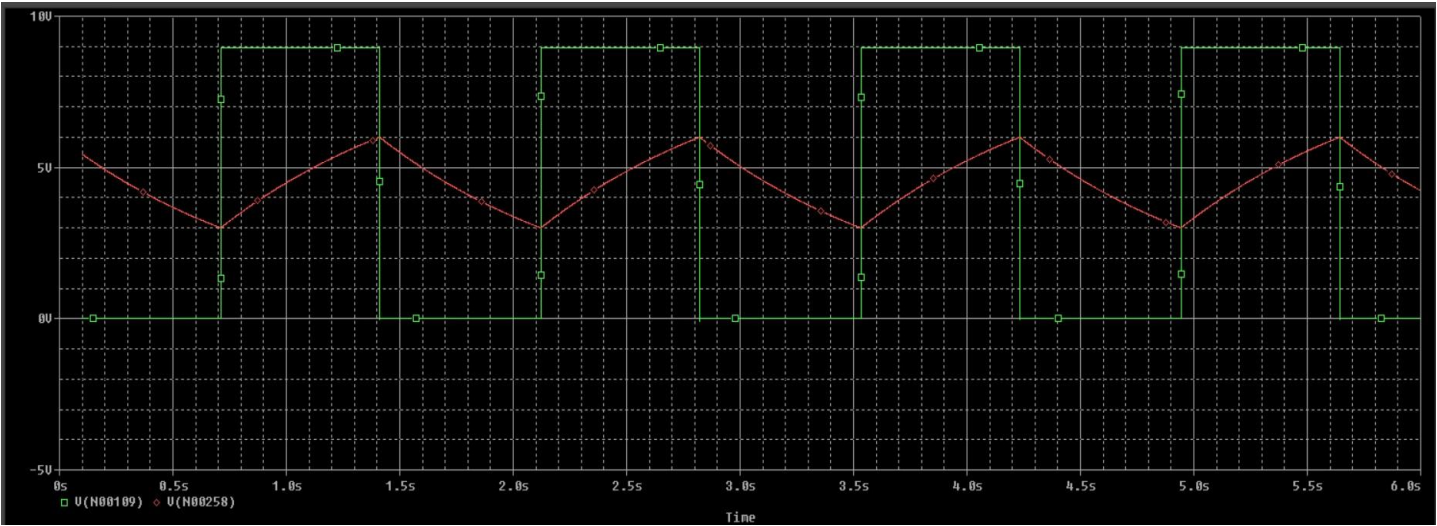


PSpice Code:

- | | |
|--|--|
| 1. * Astable multivibrator using 555C | 11. |
| 2. .lib | 12. XU1 O N2 OUT VDD N3 N2 N1 VDD 555C |
| "C:\Cadence\OrCADX_24.1\tools\pspice\Library\anl_misc.lib" | 13. |
| 3. | 14. .tran 0.1ms 200ms |
| 4. V VDD 0 DC 9V | 15. .probe |
| 5. | 16. .end |
| 6. R1 VDD N1 1k | |
| 7. R2 N1 N2 100k | |
| 8. C1 N2 0 10u | |
| 9. C2 N3 0 0.01u | |
| 10. R3 OUT 0 10k | |

Result: The 555 timer-based astable multivibrator was successfully simulated. The output waveform displayed a continuous square wave, confirming correct charging and discharging behavior of the timing capacitor and stable astable operation of the circuit.

Output Waveform:



Experiment-6

Objective: To design and simulate an RC phase shift oscillator circuit.

Theory: An RC phase shift oscillator produces a sinusoidal output using a combination of resistor–capacitor (RC) networks and an amplifying component such as a BJT or op-amp. The RC feedback network generates a total phase shift of 180° , while the amplifier provides an additional 180° , resulting in a total phase shift of 360° (or 0°), satisfying the Barkhausen criterion for sustained oscillations.

The frequency of oscillation depends on the values of the resistors and capacitors in the RC network. This oscillator is known for its simplicity and stability and is commonly used for generating audio frequencies, test signals, and waveform generation in analog systems.

Tools required:

9. OrCAD EDA Suite

10. PSpice Simulator **Procedure:**

21. Open PSpice AD and enter the required circuit code.

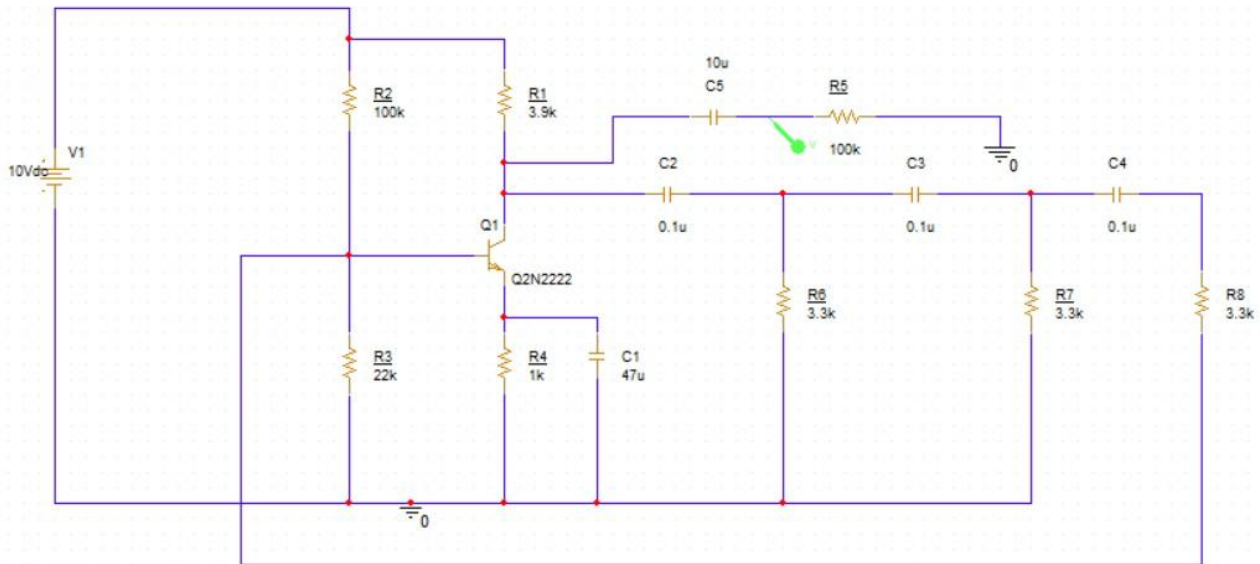
22. Save the file in “.cir” format.

23. Go to File Open Simulation and open the .cir file.

24. Click the run simulation icon at the top.

25. Add the desired trace in the output graph to view the output waveform.

Circuit Diagram:

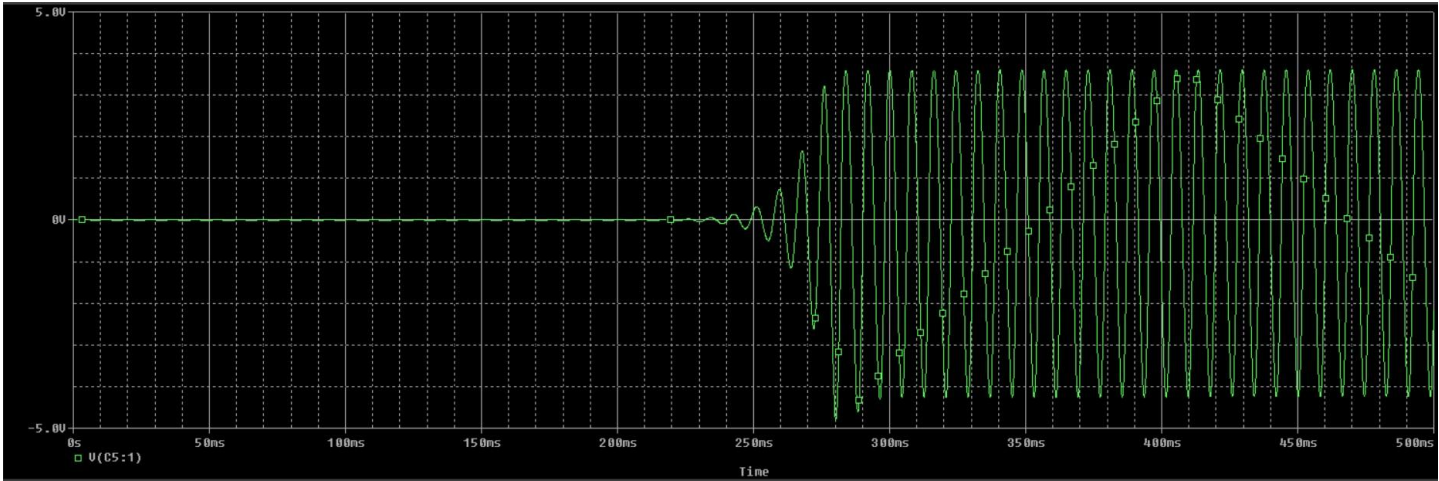


PSpice Code:

1. * RC Phase Shift Oscillator	14. C5 3 4 10u
2. V1 0 DC 10	15. R5 4 0 100k
3. R2 1 2 100k	16. Q1 3 2 5 Q2N2222
4. R1 1 3 3.9k	17.
5. R3 2 0 22k	18. .model Q2N2222 NPN (BF=200 IS=1E-14)
6. R4 5 0 1k	19. .probe
7. C1 5 0 47u	20. .tran 50u 500m
8. C2 3 6 0.1u	21.
9. R6 6 0 3.3k	22. .end
10. C3 6 7 0.1u	
11. R7 7 0 3.3k	
12. C4 7 8 0.1u	
13. R8 8 2 3.3k	

Result: The Colpitts oscillator was successfully simulated, producing a stable sinusoidal output waveform. The capacitive voltage divider provided the correct feedback required to sustain oscillations, confirming accurate frequency generation as determined by the LC network parameters.

Output Waveform:



Experiment-7

Objective: To design and simulate a Colpitts oscillator circuit.

Theory: The Colpitts oscillator is a type of LC oscillator that employs a combination of an inductor and two capacitors to form its feedback network. The capacitors act as a voltage divider, providing the necessary feedback from the output to the input to maintain continuous oscillations. The frequency of oscillation depends on the inductance of the coil and the capacitance values of the two capacitors.

This configuration is known for its stability, simplicity, and efficiency in generating high-frequency sinusoidal signals, making it widely used in RF circuits, communication systems, and signal generators.

Tools required:

11. OrCAD EDA Suite

12. PSpice Simulator **Procedure:**

26. Open PSpice AD and enter the required circuit code.

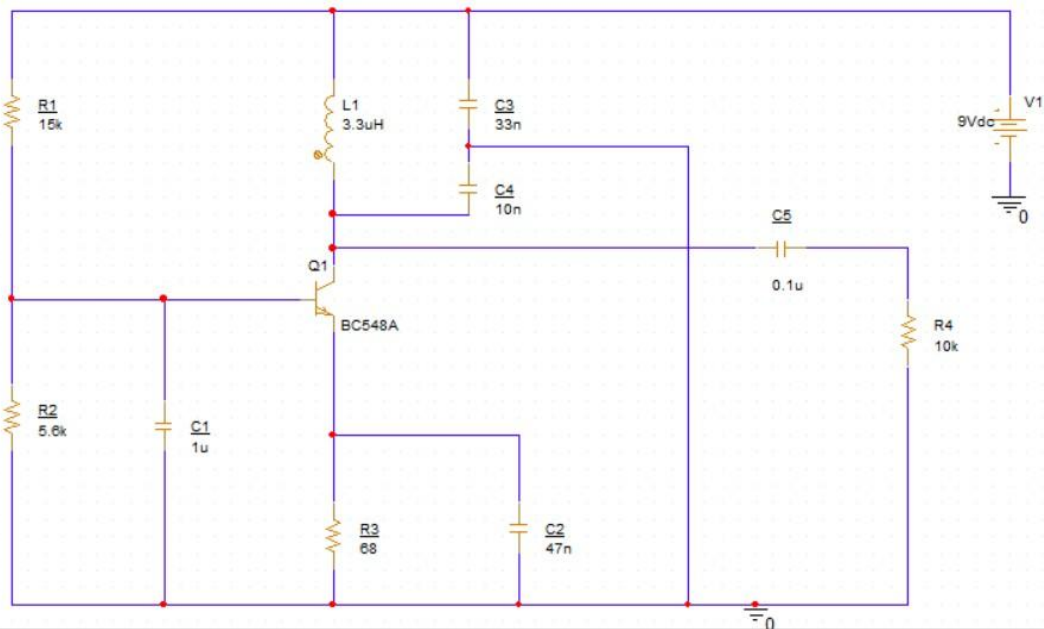
27. Save the file in “.cir” format.

28. Go to File Open Simulation and open the .cir file.

29. Click the run simulation icon at the top.

30. Add the desired trace in the output graph to view the output waveform.

Circuit Diagram:

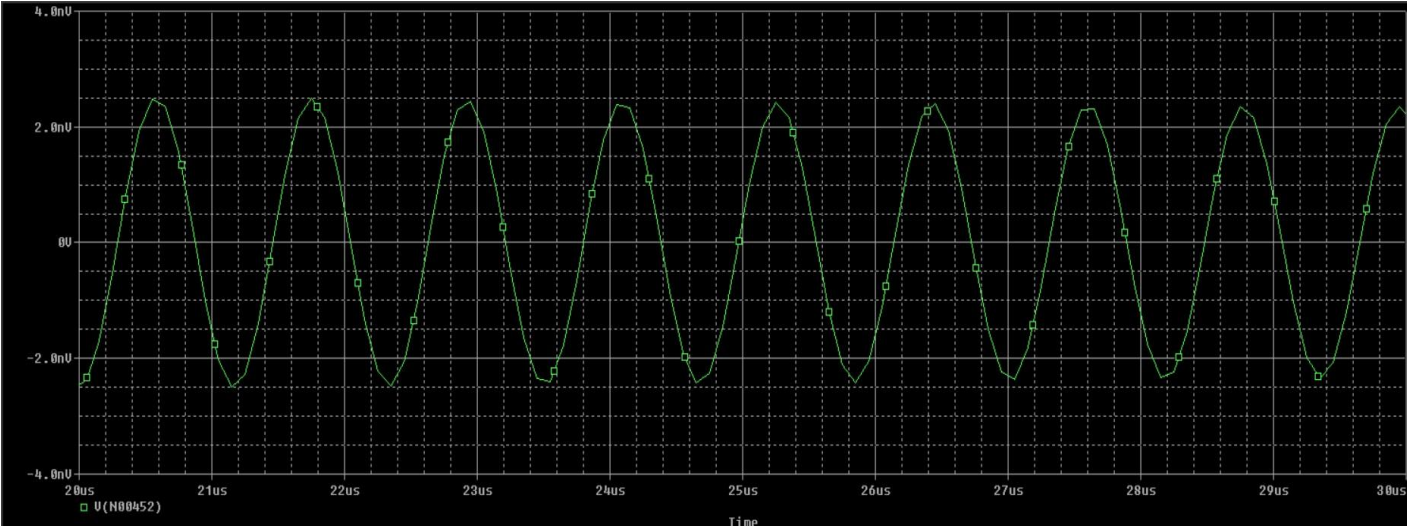


PSpice Code:

- | | |
|-------------------------------------|---------------------------------|
| 1. * Colpitt Oscillator using BC548 | 12. C5 5 2 0.1u |
| Transistor | 13. L1 2 7 3.3uH |
| 2. VCC 7 0 DC 9V | 14. |
| 3. Q1 2 3 4 BC548A | 15. .model BC548A NPN (IS=1E-14 |
| 4. R1 3 7 15k | BF=150 VAF=100 CJE=10pF CJC=5pF |
| 5. R2 0 3 5.6k | RE=1 RC=1) |
| 6. R3 0 4 68 | 16. |
| 7. R4 0 5 10k | 17. .TRAN 0 30us 0 0.1us |
| 8. C1 0 3 1u | 18. .PROBE |
| 9. C2 0 4 47n | 19. .END |
| 10. C3 0 7 33n | |
| 11. C4 2 0 10n | |

Result: The RC phase shift oscillator was successfully simulated. The output waveform displayed a continuous sinusoidal signal, confirming that the RC feedback network and amplifier together provided the required 360° phase shift and gain conditions necessary for sustained oscillations.

Output Waveform:



Experiment-8

Objective: To design and simulate a Hartley oscillator circuit.

Theory: The Hartley oscillator is an LC-based oscillator that generates sinusoidal signals using a tank circuit composed of two inductors (or a single tapped coil) and a capacitor. Feedback is derived from the center tap of the inductive network, which provides the required phase shift to maintain continuous oscillations.

The frequency of oscillation is determined by the total inductance of the two coils and the capacitance of the tank circuit. The Hartley oscillator is commonly employed in RF applications, signal generators, and oscillator stages of transmitters due to its simplicity, ease of tuning, and stable frequency output.

Tools required:

13. OrCAD EDA Suite

14. PSpice Simulator **Procedure:**

31. Open PSpice AD and enter the required circuit code.

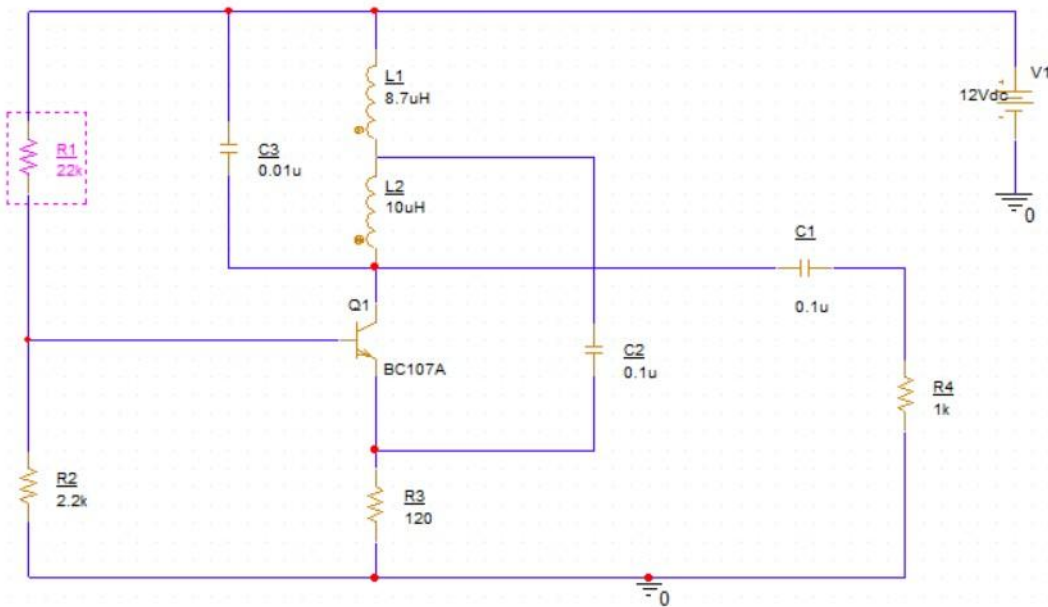
32. Save the file in “.cir” format.

33. Go to File Open Simulation and open the .cir file.

34. Click the run simulation icon at the top.

35. Add the desired trace in the output graph to view the output waveform.

Circuit Diagram:



PSpice Code:

- | | |
|------------------------------|-----------------------------------|
| 1. * HARTLEY OSCILLATOR - Q1 | 12. R4 6 0 1k |
| (BC107A) | 13. |
| 2. V1 4 0 DC 12V | 14. .MODEL BC107A NPN (IS=1.0E-14 |
| 3. R1 4 3 22k | BF=200 BR=1 CJE=8p CJC=3p |
| 4. R2 3 0 2.2k | TF=0.4n TR=50n VAF=80) |
| 5. R3 2 0 120 | 15. .TRAN 0.1u 560u 550u UIC |
| 6. Q1 5 3 2 BC107A | 16. .PROBE V(5) |
| 7. C3 4 5 0.01u | 17. .END |
| 8. L1 4 5 8.7uH | |
| 9. L2 5 2 10uH | |
| 10. C2 2 5 0.1u | |
| 11. C1 5 6 0.1u | |

Result: The Hartley oscillator was successfully simulated, producing a steady sinusoidal waveform at the output. The feedback through the inductive network ensured the required phase shift and gain conditions, confirming sustained oscillation and correct circuit operation.

Output Waveform:

